

Task 2: Report

Cucumber Disease Recognition

Introduction:

Our dataset is designed to train machine vision algorithms for the early recognition and classification of common cucumber diseases, addressing the challenge of farmers being unable to detect these issues with the naked eye.

Dataset Classes

The dataset comprises eight distinct classes of cucumber conditions:

- **Diseases (6 Classes):**
 1. Anthracnose
 2. Bacterial Wilt
 3. Belly Rot
 4. Downy Mildew
 5. Pythium Fruit Rot
 6. Gummy Stem Blight
- **Healthy Conditions (2 Classes):**
 1. Fresh leaves (presumably healthy leaves)
 2. Fresh cucumber (presumably healthy fruit)

Dataset Size and Structure

- Total Original Images: 1,280
- Images per Class: 160
 - $160 \text{ images/class} \times 8 \text{ classes} = 1,280 \text{ total images}$

Primary Goal

The dataset serves as a "state-of-the-art mentor" for developing algorithms to improve cucumber yields and utilization by providing a reliable source for training models to identify diseases early in the agricultural field.

Methodology

Multiple state-of-the-art pre-trained CNN architectures were evaluated:

ResNet50

EfficientNetB0

Vgg16

The procedure for all models include:

1. **Feature Extraction Base:** The pre-trained convolutional base was used as a fixed feature extractor (weights were frozen).
2. **Custom Head:** A custom classification head was added to the frozen base.
3. **Initial Training:** Only the top layers (the custom head) were trained initially.

Note: Hyperparameters were kept consistent across all models for fair comparison.

Results:

Performance Summary:

Model Name	Test Accuracy	Weighted F1-Score	Time(min)	Fine-Tuned?	ROC/AUC
ResNet50	0.933	0.93	96.27	Yes	Yes
EfficientNetB0	0.8299	0.8235	84.31	Yes	Yes
VGG16	0.897	0.89	203.51	Yes	Yes

Discussion

- **ResNet50 is the Top Performer:**
 - It achieved the highest Test Accuracy (0.933) and Weighted F1-Score (0.93), indicating the best overall predictive capability and class-wise balance.

- **VGG16 is the Slowest:**
 - While performing reasonably well in terms of accuracy (0.897) and F1-Score (0.89), VGG16 required the most Training Time at 203.51 minutes, making it the least efficient option computationally.
- **EfficientNetB0 is the Fastest (and Least Accurate):**
 - EfficientNetB0 had the fastest training time at 84.31 minutes. However, it recorded the lowest Test Accuracy (0.8299) and Weighted F1-Score (0.8235), suggesting a significant trade-off between speed and prediction quality for this specific task.

Conclusion:

ResNet50 offers the optimal balance of performance, delivering significantly higher accuracy and F1-score compared to the other two models. While EfficientNetB0 is the fastest, its lower performance metrics make ResNet50 the preferred model for a high-stakes application like early disease detection, where accuracy is vital.